International Journal of Pharmaceutics 637 (2023) 122839

Contents lists available at ScienceDirect

International Journal of Pharmaceutics

journal homepage: www.elsevier.com/locate/ijpharm

DE-INTERACT: A machine-learning-based predictive tool for the drug-excipient interaction study during product development—Validation through paracetamol and vanillin as a case study

Swayamprakash Patel [a,*], Mehul Patel [b], Mangesh Kulkarni [c], Mruduka S. Patel [d]

[a] Department of Pharmaceutical Technology, Ramanbhai Patel College of Pharmacy, Charotar University of Science and Technology (CHARUSAT), CHARUSAT Campus, Changa 388421, India
[b] Department of Pharmaceutical Chemistry, Ramanbhai Patel College of Pharmacy, Charotar University of Science and Technology (CHARUSAT), CHARUSAT Campus, Changa 388421, India
[c] Department of Pharmaceutical Technology, LJ Institute of Pharmacy, L J University, LJ Campus, SG Road, Ahmedabad 382210, India
[d] Research and Development (Clinical), Meteroic Biopharmaceuticals, Ahmedabad, India

ARTICLE INFO

Keywords:
Machine learning
Artificial intelligence
Drug
Excipient
Compatibility
PubChem Fingerprint

ABSTRACT

The compatibility of drugs with excipients plays a crucial role in the prospective stability of pharmaceutical formulations. Apart from real-time stability studies, conventional analytical tools like DSC, FTIR, NMR, and chromatography help identify the possibilities of drug-excipient interactions. Machine learning can assist in developing a predictive tool for drug-excipient incompatibility. In the present work, PubChem Fingerprint is employed as the descriptor of compounds that thoroughly represents the drug's and excipient's chemistry. The 881-bit binary fingerprints of each drug and excipient make 1762 inputs, and one categorical output makes an instance in the dataset. A dataset of more than 3500 instances of drugs and excipients is carefully selected from peer-reviewed research papers. Rigorous training of the Artificial Neural Network (ANN) model was performed with maximum validation accuracy, minimum validation loss, and maximum validation precision as the checkpoints. The machine learning model (DE-Interact) was trained, achieving training and validation accuracies of 0.9930 and 0.9161, respectively. The performance of the DE-Interact model was evaluated by confirming three incompatible predictions by conventional analytical tools. Paracetamol with vanillin, paracetamol with methylparaben, and brinzolamide with polyethyleneglycol are these instances which are predicted as incompatible by the DE-Interact. DSC, FTIR, HPTLC, and HPLC analysis confirm the prediction. The present work offers a reliable DE-Interact tool for quick referencing while selecting excipients in formulation design.

1. Introduction

1.1. Drug excipient interaction

One key parameter that needs to be reviewed during pharmaceutical formulation development is the drugs and excipients compatibility. However, while excipients play a pivotal role in the formulation's performance, their interactions with the drug may affect the overall product performance and alter the product's therapeutic effect (Akers, 2002; García-Arieta, 2014; Narang et al., 2012; Panakanti and Narang, 2012; Zarmpi et al., 2020).

Accelerated or real-time stability studies provide insight into possible drug-excipient interactions. According to ICH guidelines, real-time stability studies need to be performed for at least 12 months, while accelerated stability studies need to be performed for at least six months. (Bajaj et al., 2012; González-González et al., 2022; Pokharana et al., 2018). However, this approach can be tedious and time-consuming as a preliminary study during formulation development. Setting up real-time or accelerated stability studies for various drug-excipient combinations at the product development stage can extend the timeline.

Apart from stability testing, pharmaceutical formulators utilize many thermo-analytical and spectroscopic approaches to identify drug-excipient interactions. Differential scanning calorimetry (DSC), hot

* Corresponding author at: Department of Pharmaceutics, Ramanbhai Patel College of Pharmacy, Charotar University of Science and Technology (CHARUSAT), CHARUSAT Campus, Changa 388421, India.
E-mail addresses: Swayamprakash.patel@gmail.com, Swayamprakashpatel.ph@charusat.ac.in (S. Patel).

https://doi.org/10.1016/j.ijpharm.2023.122839
Received 15 December 2022; Received in revised form 4 March 2023; Accepted 10 March 2023
Available online 15 March 2023
0378-5173/© 2023 Elsevier B.V. All rights reserved.

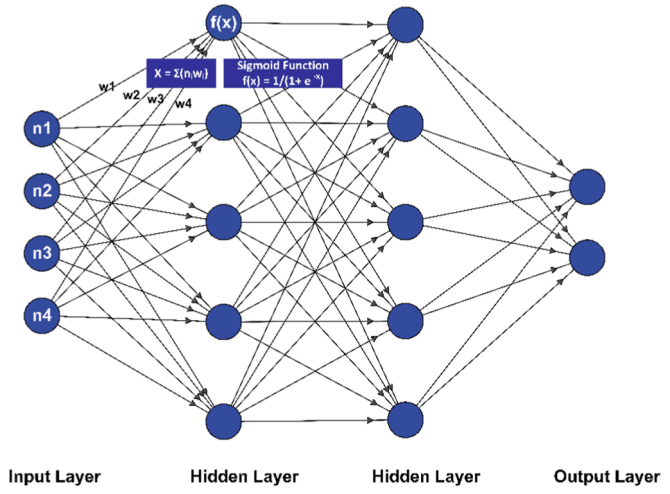


Fig. 1. Basic architecture of ANN.

stage microscopy (HSM), powder X-ray diffraction (PXRD), solid-state NMR, scanning electron microscopy (SEM), and high-performance chromatography can be employed for incompatibility detection between the drug and excipient (Chadha and Bhandari, 2014a; Thomas and Naath, 2008; Wakasawa et al., 2008). These methods are pretty effective, but they require sophisticated and costly equipment. As a result, aspirants often have to rely on public or private testing facilities.

Artificial intelligence (AI) and machine learning (ML) based techniques can identify hidden patterns within complex systems. Based on the features extracted from a large dataset, the ML model can recognize the presence of functional groups that are possibly responsible for drug-excipient interactions and can classify pairs of drugs and excipients as either compatible or incompatible accordingly.

1.2. Artificial neural network (ANN)

ANN is a crucial part of artificial intelligence. It is a computational system developed by considering the neural network of the human brain (Agatonovic-Kustrin and Beresford, 2000). The human brain takes inputs from various sensory organs, such as the eyes, ears, and touch receptors. Every information received as the input gets processed in a complex network of neurons in the brain, which results in a specific output at the end. The brain makes decisions based on its past experiences and training. For instance, it knows what output to generate in case of accidental contact with a hot surface, as it has been trained to move away from the hot surface immediately (Hassoun, 1995).

Similarly, an ANN consists of a complex structure of neurons where complex mathematical calculations occur, resulting in the desired output. Once trained with appropriate data, an ANN can be employed to make proper decisions in an artificial intelligence system (Maind and Wankar, 2014; Rumelhart et al., 1994).

As shown in Fig. 1, ANN has three layers: a) the Input layer, b) the Hidden layer, and c) the Output layer (Gupta, 2013). Each layer consists of nodes with specific values (n_i) carried forward to the next layer. Each layer consists of nodes. Nodes are some values (n_i) carried forward to the next layer. Each node is interconnected with the nodes of the next layer. This connection represented here with arrows is nothing but random values known as weight (w_i). Values of input layers are multiplied with weights and presented to the respective node of the hidden layer. Within the hidden layer, various mathematical functions are employed to this value. The results of this function are carried forward to the next layer as the input. During the training of ANN for a given input dataset, weights are changed randomly by the specific algorithm until output with minimum error is achieved. Once ANN is trained with maximum accuracy, it can be used for prediction for unknown inputs. The detailed discussion of ANN is out of the scope of the present work. Refer to ANN from the literature elsewhere for a better understanding (Agatonovic-Kustrin and Beresford, 2000; Gupta, 2013; Han et al., 2018; Hassoun, 1995).

Table 1
Different types of Fingerprints which the Pa-DEL descriptor can calculate.

Type of Fingerprints	Number of Bits	Description
CDK fingerprint	1024	The fingerprint length is 1024, and the search depth of 8
CDK extended fingerprint	1024	Extends the Fingerprinter with additional bits describing ring features
Estate fingerprint	79	E-State fragments
CDK graph only fingerprint	1024	A specialized version of the Fingerprinter which does not take bond orders into account
MACCS fingerprint	166	MACCS keys
PubChem fingerprint	881	PubChem fingerprint
Substructure fingerprint	307	Presence of SMARTS Patterns for Functional Group Classification by Christian Laggner
Substructure fingerprint count	307	Count of SMARTS Patterns for Functional Group Classification by Christian Laggner
Klekota-Roth fingerprint	4860	Presence of chemical substructures
Klekota-Roth fingerprint count	4860	Count of chemical substructures
2D atom pairs	780	Presence of atom pairs at various topological distances
2D atom pairs count	780	Count of atom pairs at various topological distances

Table 2
Details of PubChem Fingerprint.

Sections	Bit Positions	Fingerprint Description
Section 1	0–114	Hierarchic Element Counts
Section 2	115–262	Rings in a canonic Extended Smallest Set of Smallest Rings (ESSSR) ring set
Section 3	263–326	Simple atom pairs
Section 4	327–415	Simple atoms nearest neighbours
Section 5	416–459	Detailed atom neighbourhoods
Section 6	460–712	Simple SMARTS patterns
Section 7	713–880	Complex SMARTS patterns

1.3. PubChem Fingerprint

Molecular descriptors play a pivotal role in the evolution of computational chemistry, chemoinformatics, and artificial intelligence in drug discovery (Grisoni et al., 2018; Xue and Bajorath, 2000). These descriptors represent various functional and structural properties of molecules, such as molecular weight, molar refractivity, logarithmic partition coefficient (LogP), number of acidic groups, etcetera. The PaDEL descriptor is an open-source tool for molecular descriptor calculation (Yap, 2011), which can calculate 1875 types of descriptors and 12 different types of fingerprints. To generate all these descriptors, the PaDEL descriptor requires input details of molecules in the structure-data file format (.sdf).

A fingerprint is a complex but comprehensive molecule representation, generally presented as a string of binary bits. Each bit represents the presence or absence of a particular molecular functional or structural segment. Table 1 shows the details for the different types of fingerprints generated through the PaDEL descriptor.

The PubChem fingerprint is an 881-bit-long binary string consisting of seven different sections. These sections consist of several bits and occupy specific bit positions, respectively. As shown in Table 2, each section represents a structural component of the molecule, with one and zero at particular bit positions indicating the presence and absence of a

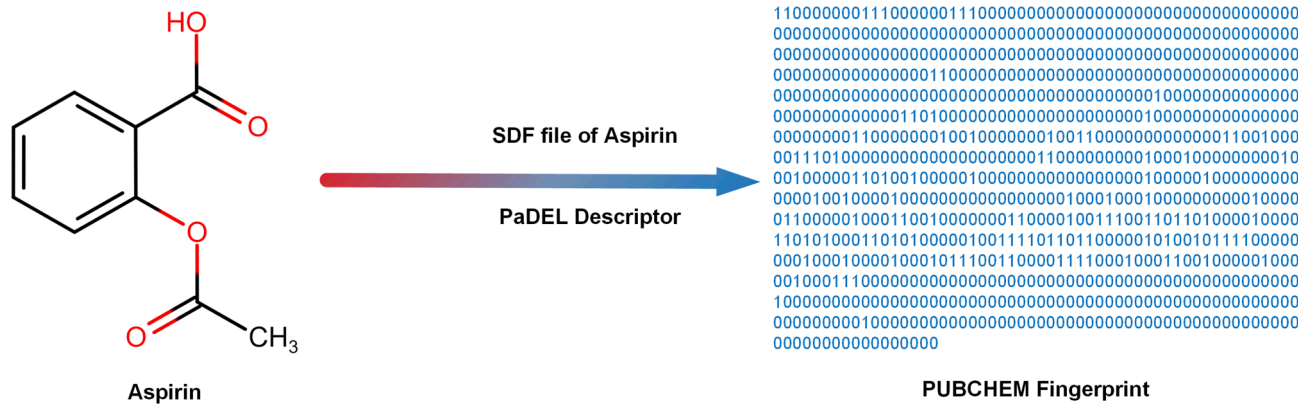


Fig. 2. Calculation of PubChem Fingerprint using PaDEL Descriptor tool. PubChem Fingerprint is an 881-bit long binary string representing seven different structural components of the molecule.

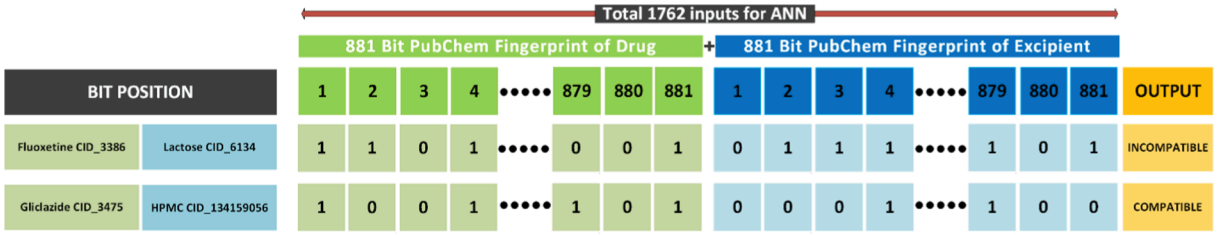


Fig. 3. Collection and compilation of data for drug-excipient.

particular structural component, respectively. Fig. 2 depicts an example of the PubChem fingerprint for aspirin. More details regarding each bit position can be found on the PubChem website.

The PubChem fingerprint represents elements, types of ring systems, atom pairing, atom environment, and other molecule characteristics. As the PubChem fingerprint represents the presence or absence of various structural components in a molecule, it is ideal for developing AI-based predictive tools for possible interactions between drugs and excipients. Structural components in drugs and excipients are responsible for possible interactions between them. Additionally, the PubChem platform is developed and maintained under the National Library of Medicine (NLM) project of the US Government, which ensures the continuous update, upgrade, and maintenance of molecular fingerprinting technology. It is one of the reasons for selecting the PubChem fingerprint for the current project over other options.

In the present work, an ANN-based predictive model is developed to identify drug-excipient interaction. PubChem fingerprints of the drug and excipient are utilized as the input, while "Compatible" and "Incompatible" are selected as two output class labels.

2. Method

2.1. Dataset preparation

Artificial neural network-based (ANN) predictive model requires input and output datasets for training, testing and validation. For the present work, we utilized the PubChem fingerprint of the drug and excipient as input and their respective chemical interaction as output.

Data for drug-excipient incompatibility were collected from peer-reviewed research publications. Each drug's structure-data files (.SDF) and excipients were collected individually from PubChem. The PubChem fingerprints were generated using the PaDel-descriptor tool for drug and excipient (Yap, 2011). As shown in Fig. 3, for each pair of drugs and excipients, their respective fingerprints were appended in Microsoft Excel, followed by their type of interaction (incompatible or compatible) as an output. So, the data consist of 1763 columns, from which 1762 are input columns, and one is a target column.

The interaction details of more than 3500 combinations of drug and excipient were collected from peer-reviewed research papers (Aysha et al., 2019; Chandrakala et al., 2011; Gaikwad et al., 2016; Ghaferi et al., 2020; Gorain et al., 2018; Israr et al., 2014; Liang, 2014; Nivedithaa and Saba, 2018; Remya et al., 2010; Zhou et al., 2017) and the USFDA database for drugs. From the USFDA database for drugs, we have selected the label of reference listed drugs (RLD). Excipients listed on the label of RLD were considered compatible combinations for particular drugs.

2.2. Modelling and characterization protocol

2.2.1. Data preprocessing

A supervised learning model requires a labelled dataset. The input feature that comprised two concatenated fingerprints (drug and excipient) was labelled manually as either "Compatible" or "Incompatible" (Refer: Fig. 3). It made the categorial output feature with two different categories. The one-hot encoding was employed to transform categorical data in the numerical representation of [1,0] and [0,1] for the "Compatible" and "Incompatible" categories, respectively. The One-Hot Encoder from the sci-kit-learn library was employed for the one-hot encoding of the output.

2.2.2. ANN model

A feed-forward ANN model was trained with different architectures, including varying numbers of hidden layers, nodes, and activation functions. The best architecture on the training set was selected and presented herewith.

The 881-bit binary fingerprint of the drug and excipient makes 1762 nodes in the input layer. At the same time, the single node (two nodes after one-hot encoding) of the output layer classifies the data in either compatible or incompatible categories. Three hidden layers with 1024 nodes in each, for the 1762 nodes in the input layer, were employed as a result of several trials for maximum possible accuracy, precision, and F1-score. Rectified Linear (ReLu) functions for hidden layers and Softmax

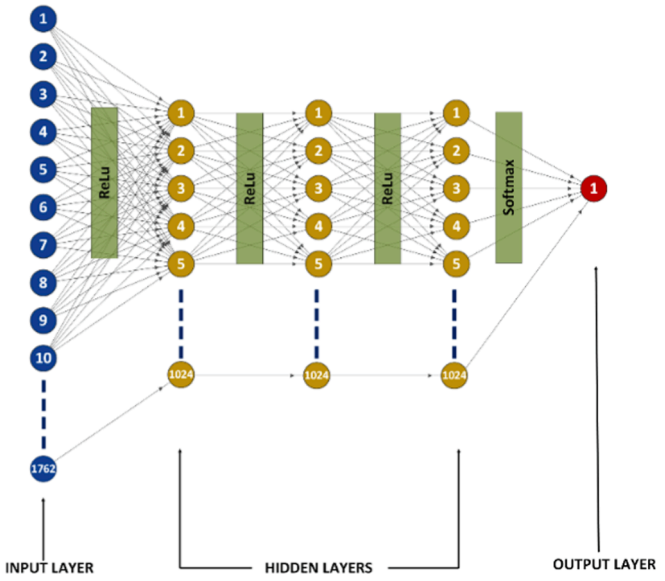


Fig. 4. ANN architecture for Drug-Excipient Interaction Prediction.

function for one-hot vectored output layer was utilized, as shown in Fig. 4. The ANN models were trained with a maximum of 200 epochs and a batch size of 150.

The data were split into the train (70%), test (15%) and validation (15%) set using Scikit-learn open-source library (Garreta and Moncecchi, 2013; Pedregosa et al., 2011). The entire work, including the dataset and codes, can be accessed from the GitHub repository.

2.2.3. Model evaluation

The root means square propagation (RMSprop) algorithm (Kumar et al., 2020; Labhsetwar et al., 2020; Yaqub et al., 2020) as an optimizer and binary cross entropy with logits (Ruby et al., 2020) as a loss function were implemented. ANN model was trained separately at three different checkpoints: a) maximum validation accuracy, b) minimum validation loss and c) maximum validation precision.

All three models' parameters were compared to select the best ANN model. The maximum validation precision checkpoint was performed to justify the skewness in the dataset. For the training of this ANN model, more than 90% of the training dataset was employed for validation. Both oversampling and K-Fold cross-validations (with 5, 7 and 10 folds) techniques were employed to understand the effect of imbalanced data and to select the best model. The model with the best accuracy, precision and F1-score indicators was selected as the final DE_Interact predictor.

2.3. PubChem Integration

The PubChemPy API (https://pubchempy.readthedocs.io/en/latest/index.html) was employed for integrating the trained ANN model with the PubChem database for the prediction of drug-excipient interaction using the PubChem Compound ID (CID) (Swain, 2017). ANN does not discriminate between the drug and the excipient as it is based on the fingerprint. A graphical user interface was developed using Google Colab® wherein two separate fields for the drug and excipient were set for the prediction of interaction using the trained ANN model (DE-Interact).

2.4. Validation of DE_Interact model

Numerous drug and excipient combinations (not part of the training or test dataset) are screened using the DE_Interact model. The model predicted a combination of paracetamol and vanillin as an incompatible combination. The model's prediction was validated by employing various qualitative and quantitative analyses.

Differential Scanning Calorimetry (DSC), Fourier Transform Infrared (FTIR) spectroscopy and High-Performance Thine Layer Chromatography (HPTLC) analysis and High-Performance Liquid Chromatography (HPLC) studies were performed for the paracetamol and vanillin.

Samples of Paracetamol, vanillin and their physical mixture in equal proportion were analyzed for incompatibility. Fresh samples and samples stored in a dry place at room temperature for at least one month were analyzed.

2.4.1. Differential scanning calorimetry (DSC)

DSC is considered a quick check of drug-excipient incompatibility (Balestrieri et al., 1996; Chadha and Bhandari, 2014b). The significant peak height and position change may detect possible interaction between the two compounds. DSC was performed using Mettler Toledo's model TGA-DSC-1 as a combined unit of TGA and DSC. DSC measurements for paracetamol, vanillin, and physical mixture (1:1) were performed within the temperature range of 25 oC to 250 oC and at a heating rate of 5 °C per minute without the nitrogen purge.

2.4.2. Fourier Transform Infrared spectroscopy (FTIR)

All powdered drug samples, excipients and drug-excipient combinations were scanned using a Shimadzu Fourier Transform Infrared Spectrophotometer IRSpirit® Series QATR-S with a scan range between 4000 and 400 cm-^{1}. % Transmittance was implemented as the intensity mode. The instrument was calibrated to a resolution of 4 cm^{-1}. For post-run, LabSolutionsIR® was used. The sample powders were prepared prior to the ATR-FTIR scan. FTIR analysis of the physical mixture (1:1) of paracetamol and vanillin was performed for the fresh and old sample, which was stored at room temperature for one month. Powdered samples were placed on the ATR platform under the diamond and analyzed at the scanning range of 4000 – 400 cm^{-1}.

2.4.3. High-Performance thin layer chromatography

The instability of paracetamol in the presence of vanillin was further confirmed with the help of high-performance thin-layer chromatography (HPTLC). The chromatograms of paracetamol, vanillin and their physical mixture (1:1) were developed for the fresh and one-week-old samples stored at room temperature in the tightly closed Eppendorf tubes.

The methanolic samples (2.0 µl) were sprayed in a band pattern over the aluminium HPTLC silica gel 60F_{254} plates as a stationary phase using the CAMAG® Linomat5 semi-automatic sample application system. HPTLC plates were developed in a pre-saturated TLC chamber containing mobile phase (Chloroform (8 ml), Acetone (1.9 ml), Ammonia (0.1 ml)). Each sample's chromatogram was developed using the CAMAG® TLC scanner at 243 nm λ_{max} and analyzed using the chromatography managing tool – winCATS® version 1.4.7.

2.4.4. High-pressure liquid chromatography

The content of paracetamol was determined in the fresh and one-week-old physical mixture (1:1) of paracetamol and vanillin using RP-HPLC. Inertsil® ODS 3 V (150 × 4.6 mm, 5 µm) was used as the stationary phase, while the mixture of trifluoroacetic acid (0.1 % v/v) and acetonitrile (70:30) was employed as the mobile phase. Chromatogram was recorded at a flow rate of 1 ml/min at 257 and 274 nm wavelengths.

3. Result and discussion

3.1. Data

It is indeed cumbersome to collect the data for incompatible combinations. The dataset for the current project consists of 179 incompatible combinations and 3549 compatible combinations of drugs and excipients. The dataset seems skewed, wherein the compatible combinations are more heavily weighted than incompatible combinations.

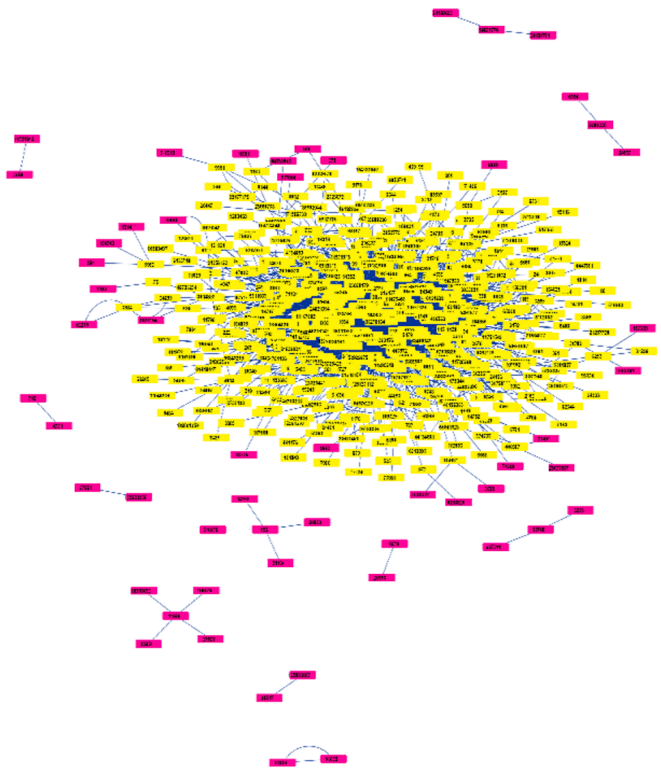


Fig. 5. Network graph for the combinations of drug and excipient (Nodes with at least five neighbours with at least two distances are highlighted yellow.).

However, the quality of data is not compromised. The network graph for the dataset, as shown in Fig. 5, represents the presence of multiple combinations of drugs and excipients. The network statistics represent 739 nodes (3728 edges) with a 9.879 average number of neighbours. So, 739 drugs and excipients are involved in the entire dataset. Nodes with at least five neighbours are highlighted with yellow in Fig. 5, which embrace a significant network area. It reveals the richness of data.

The data quality is quite good, but it is statistically imbalanced. The label "Compatible" contributes more than 95% (n = 3549), and the label "Incompatible" covers only 4.80% (n = 179) of the total dataset respectively. The impact of the imbalanced dataset is minimized through oversampling incompatible data in the training and validation dataset (Bach et al., 2017; Ertekin, 2013; Mathew et al., 2017). Oversampling is performed by replicating the data for the "Incompatible" class four times. Thus, the newer dataset contains 83% of the compatible data class and 16.38% of the incompatible class data. The prediction of the incompatible class is more critical than compatible. Comparatively, a lower percentage of incompatible classes in the dataset makes the prediction even more reliable if it is predicting the incompatible class.

However, apart from the oversampling, the model was also trained by employing the stratified K-Fold cross-validation approach to tackle an imbalanced dataset. The neural network model was also trained by implementing the 5, 7, 10 folds cross-validation. None of the techniques (oversampling and stratified K-fold) showed any significant difference in the accuracy, precision and F1 score. So, the oversampling technique was finalized.

3.2. ANN model

ANN models were trained separately at two different checkpoints for monitoring the maximum validation accuracy and the minimum validation loss for the accuracy as the performance metrics. Fig. 6 and Fig. 7 depict the logistic change in accuracy and loss with epochs for the maximum validation accuracy and minimum validation loss as the checkpoints, respectively. While Fig. 8 shows the precision and loss versus epochs for the checkpoint set to maximum validation precision.

For maximum validation accuracy as the checkpoint, as shown in Fig. 6, maximum training accuracy (0.9930) was achieved after 25 to 30 epochs. Nevertheless, the ANN training was continued up to the 200th epoch, and the weights were saved once maximum validation accuracy (0.9161) was achieved. Similarly, the minimum training validation loss (0.0115) was achieved after 25 odd epochs for minimum validation loss as the checkpoint. Weights were secured once the minimum validation loss (0.2784) was achieved. Minimum training & validation losses and maximum training & validation accuracies for the ANN model's training at two different checkpoints are shown in Table 3. ANN model with the maximum validation precision checkpoint shows even better training and validation precision and loss (Table 3) than the previous two models. The training was continued up to the 200th epoch.

Table 4 and Table 5 illustrate the confusion matrix results and class-wise value of various performance indexes like precision, recall and F1 score. Confusion matrices and performance matrices for ANN models developed with three different checkpoints are calculated. The full dataset was utilized for the calculation of confusion and performance matrices. Accuracy, precision, recall and F1 scores represent the performance of various models. Though accuracy is a straightforward representation of the model performance, it is unsuitable for imbalanced data, such as in the current scenario. For the problem of drug and excipient incompatibility, it is more important to get true incompatible than true compatible. Precision and recall are more important than accuracy for imbalanced data. A careful observation of Table 4 shows that the false positive (wrongly predicted as compatible) observations are 12 for maximum validation accuracy and maximum validation precision, while it is 36 for the case of the minimum validation loss model, respectively. So based on recall, as a parameter (True Positive / (True

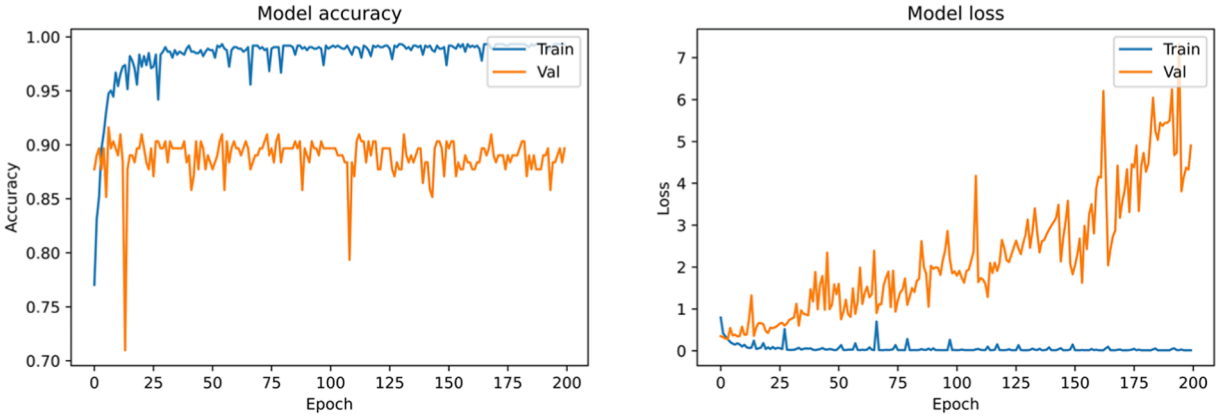


Fig. 6. Accuracy and Loss vs Epoch graph for training and validation dataset; (Checkpoint set to maximum validation accuracy).

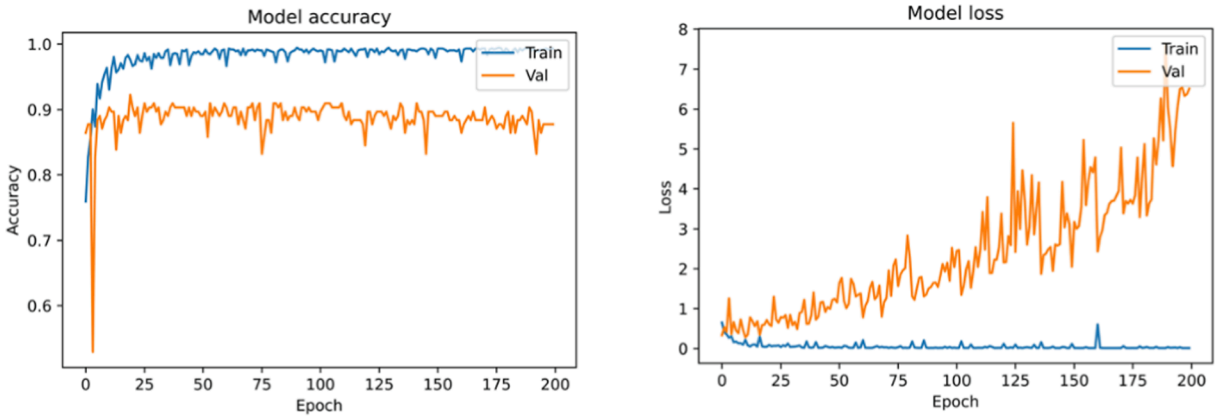


Fig. 7. Accuracy and Loss vs Epoch graph for training and validation dataset; (Checkpoint set to minimum validation loss).

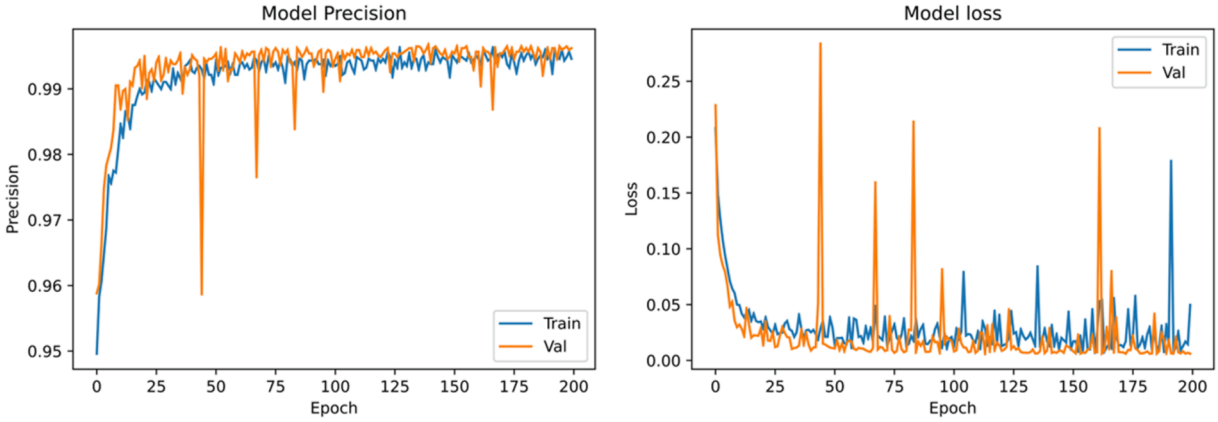


Fig. 8. Precision and Loss vs Epoch graph for training and validation dataset; (Checkpoint set to maximum validation precision).

Table 3
Parameters of trained ANN models (Max. Training & Validation accuracies or precision and minimum training & validation loss during ANN training).

Parameters	Checkpoints of ANN Models		
	Max. Validation Accuracy	Mini. Validation Loss	Max. Validation Precision
Training Accuracy or Precision	0.9930	0.9944	0.9967
Validation Accuracy or Precision	0.9161	0.9225	0.9964
Training Loss	0.0114	0.0115	0.0078
Validation Loss	0.2762	0.2784	0.0061

Positive + False Negative)), the minimum validation loss rules out explicitly. In the first stance, based on precision (kTrue Positive / (True Positive + False Positive)), the maximum validation accuracy seems better, as per Table 4 and Table 5. However, as shown in Table 3, the accuracy and precision indexes are far better during the model developed for maximum precision validation checkpoint. The F1 score also seems better for the maximum validation accuracy model, as shown in Table 5. In the present scenario of the imbalanced dataset and more emphasis on the true incompatible prediction, the recall is more important than the precision. In the F1 score, the β value is usually considered one by assuming equal importance of precision and recall. F_β value (with $\beta = 2$ for more importance to Recall (Flach and Kull, 2015; Lee et al., 2021; Shendryk et al., 2018)) is manually calculated, which are 0.975 and 0.967 for maximum validation accuracy and maximum validation precision models, respectively. So F_β is equal for both models. Conclusively, the maximum precision model seems to be performing better.

All values in performance matrices of all three ANN models are more than 0.9, which is satisfactory. Moreover, the F-statistics (F_{Crit} 5.14, p-value 0.68) among various training and validation parameters (Table 3) do not represent any significant difference among the three models. However, the precision matrix is preferable in case of imbalanced data (Chawla, 2010). So, the ANN model (AUC = 0.9876) validated based on maximum precision is finalized for further wet lab-based validation.

Table 4
Confusion matrices of predicted and actual outputs for ANN models trained at various checkpoints.

Confusion Matrix	Checkpoints of ANN Models					
	Max. validation accuracy		Min. validation loss		Max. validation precision	
	Predicted Compatible	Predicted Incompatible	Predicted Compatible	Predicted Incompatible	Predicated Compatible	Predicted Incompatible
Actual Compatible	3518	31	3530	19	3493	56
Actual Incompatible	12	687	36	663	12	687

Table 5
Classification report of ANN model trained at various checkpoints.

Class	Checkpoints of ANN Models									
	Max. validation accuracy			Min. validation loss			Max. precision			Support
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	
Compatible	1.00	0.99	0.99	0.99	0.99	0.99	1.0	0.98	0.99	3549
Incompatible	0.96	0.98	0.97	0.97	0.95	0.96	0.92	0.98	0.95	699

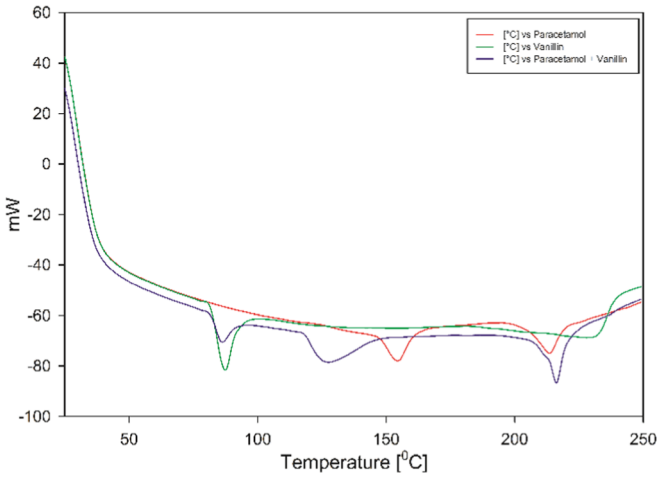


Fig. 9. DSC analysis of PCM-Vanillin Mixture.

3.3. Validation of DE-Interact

Paracetamol and vanillin were selected as unique out of many combinations predicted as incompatible by the DE-Interact model. DSC, FTIR, HPTLC, and HPLC analysis points to the incompatibility of this combination.

3.3.1. Differential scanning calorimetry (DSC)

Fig. 9 represents the DSC results of pure paracetamol, pure vanillin, and a freshly prepared physical mixture (1:1) of paracetamol and vanillin. The reported melting point of paracetamol (169 °C) can be confirmed from the DSC of Paracetamol. The same for vanillin (80 – 83oC) is also confirmed in the DSC of vanillin. The conflicts at the peak of paracetamol are conspicuously observable for the mixture. The peak is shifted to the left (around 125 °C) and broadens significantly. The peak of vanillin is undisturbed and exactly overlain with the peak of pure vanillin. This mismatch of peaks is a sign of thermodynamic instability or PCM interaction in vanillin's presence. These results support the prediction of DE-Interact.

3.3.2. Fourier Transform Infrared spectroscopy (FTIR)

FTIR study reveals the visible changes in the spectra of a fresh and one-month-old mixture of paracetamol and vanillin. The characteristic split peak of paracetamol in the region of frequencies from 3327 cm^{-1} to 3000 cm^{-1} (see standard FTIR spectra as per ChemSpider® ID: 1906 for Acetaminophen, NIST Standard Reference Database 35) is visible in the fresh mixture of paracetamol and vanillin. The narrow peak from 3327 cm^{-1} to 3324 cm^{-1} are characteristic vibrational peaks for NH. While broader bands with maxima at 3205 cm^{-1} and 3161 cm^{-1} represent the OH stretching (Burgina et al., 2004). Surprisingly, these split peaks disappeared in the FTIR spectra of a one-month-old mixture of paracetamol and vanillin. The results are shown in Fig. 10. The sharp peak of NH is missing in the old mixture of paracetamol and vanillin. It may be due to the interaction between the amide and hydroxyl groups of paracetamol and vanillin. This interaction may lead to the formation of imine, which might be the reason behind the vanishing of the sharp stretching peak of NH. However, it can only be confirmed with other complex analytical tools like mass spectroscopy and NMR. FTIR spectra here confirm the possible interaction between paracetamol and vanillin. FTIR spectra support the prediction of DE-Interact, too.

3.3.3. High-performance thin layer chromatography

The data points of chromatograms of HPTLC were exported and plotted in juxtaposition, presented in Fig. 11. The details of the peaks are summarized in Table 6.

The chromatographic parameters indicate the significant quantitative deduction of PCM in the old mixture. All samples observed a slight variation in peak heights and peak areas between fresh and old samples. However, the peak area ratio of PCM to vanillin overtly presents a significant reduction in the old mixture compared to the fresh one. The graphical presentation in (Fig. 11) can comprehend the same. In short,

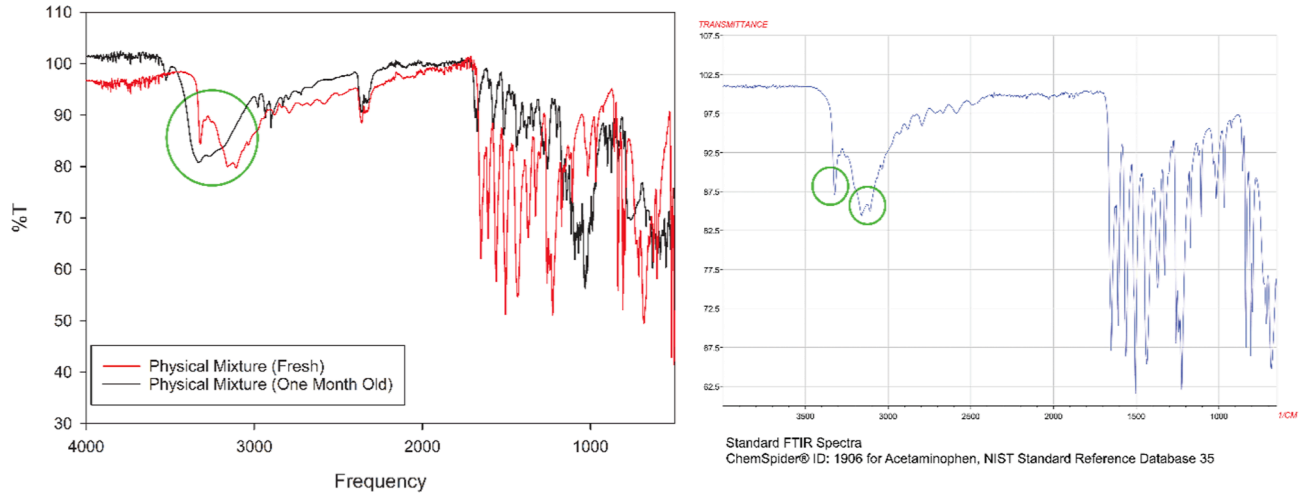


Fig. 10. FTIR study of PCM-Vanillin physical mixture and Standard FTIR spectra of paracetamol.

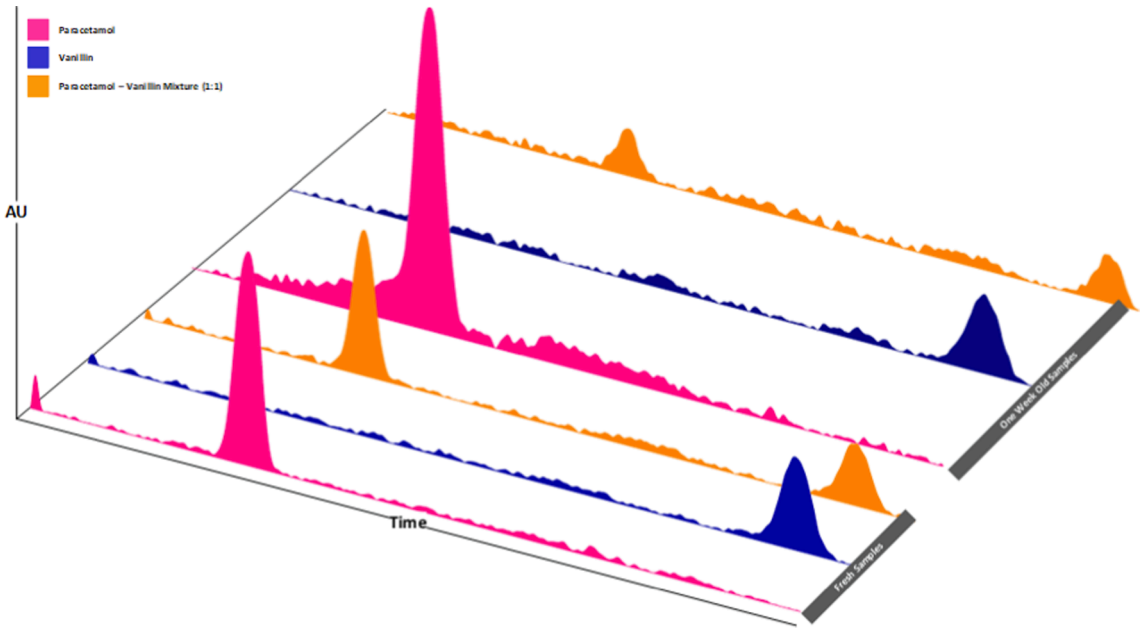


Fig. 11. HPTLC chromatogram of fresh and old samples of paracetamol, vanillin, and their physical mixture.

Table 6
Chromatographic parameters of HPTLC analysis.

Sample	Fresh	One Week Old	Parameter
Paracetamol	453.4	424.0	Peak Height (AU)
	11459.3	12416.1	Peak Area (AU)
	0.20	0.23	Rf
Vanillin	188.8	105.4	Peak Height (AU)
	5770.6	3577.2	Peak Area (AU)
	0.74	0.75	Rf
Paracetamol and Vanillin Physical Mixture (1:1)	307.2 (PCM) 123.8 (Vanillin)	58.6 (PCM) 59.2 (Vanillin)	Peak Height (AU)
	2.35	0.98	Peak Height Ratio (PCM: Vanillin)
	7075.9 (PCM) 3600.6 (Vanillin)	1326.0 (PCM) 1539.4 (Vanillin)	Peak Area (AU)
	1.965	0.86	Peak Area Ratio (PCM: Vanillin)
	0.20 (PCM) 0.75 (Vanillin)	0.25 (PCM) 0.80 (Vanillin)	Rf

HPTLC analysis also abides by the prediction of DE-Interact. PCM and vanillin show weird interactions.

3.3.4. *High-Pressure Liquid chromatography (HPLC)*

HPLC analysis was performed to determine quantitative fluctuation in the mixture of PCM and vanillin upon storage. The first peak at around Rf of 4.8 min represents the peak of PCM. The results of HPLC analysis do not show any quantitative deduction in either PCM or vanillin upon a superficial observation. However, contour analysis shows the chaos in the peak of PCM with the old sample. In the case of PCM, the peak distortion can also be observed in the HPLC chromatogram of the old mixture (Fig. 12). These results indeed affirm the interaction between PCM and vanillin. This interaction may not contribute to the quantitative determination. However, distortion in the HPLC peak of PCM indicates the physicochemical changes in PCM during the storage in the presence of vanillin. The findings of HPLC analysis also align with the prediction of DE-Interact.

DSC, FTIR, HPTLC and HPLC confirm the interaction between paracetamol and vanillin. These results certainly validate the prediction of DE-Interact. Other incompatible predictions by DE-Interacts, such as paracetamol with methylparaben and brinzolamide with polyethyleneglycol, were also validated through either DSC FTIR or morphological studies. The characteristic peaks representing the melting point of paracetamol and brinzolamide in the DSC thermogram disappeared or distorted in the presence of their excipient counterpart, respectively (Fig. 13 Left).

FTIR study for the physical mixture (1:1) of paracetamol and methylparaben as the fresh and one-month-old sample (stored at room temperature) shows complete disappearance of the characteristic peak of methylparaben (C = O and CH_3) at 1740 cm^{-1} and 2956 cm^{-1} respectively (Fig. 14). It suggests possible hydrolysis and decarboxylation of methylparaben in the presence of paracetamol. Such changes are not observed in the samples of methylparaben as alone.

Fig. 13 (Right) clearly reasserts the DE-Interact prediction for brinzolamide and polyethyleneglycol. The peak of brinzolamide at its melting point of 131 °C is wholly distorted and everted in the case of the physical mixture of brinzolamide with PEG 8000. The melting point of PEG 8000 was found unaltered at 55 °C to 60 °C in the same mixture. All grades of PEG give incompatibility with the brinzolamide. Morphological examination shows explicit colour change within 24 h in the case of PEG 400 (Fig. 15). A significant loss in the characteristic FTIR peak of brinzolamide at 1740^{-1}cm (Fig. 15) also suggests a strong possibility of incompatibility between polyethyleneglycols.

Three instances are not enough to concretely confirm the performance of DE-Interact. However, as the validation is a continuous process, no specific number of instances can complete the model's validity. Including these confirmed incompatibilities in the dataset may enhance the model's predictive accuracy.

Other incompatible combinations of the drug-excipient predicted by the DE-Interact, which are neither part of our dataset nor reported

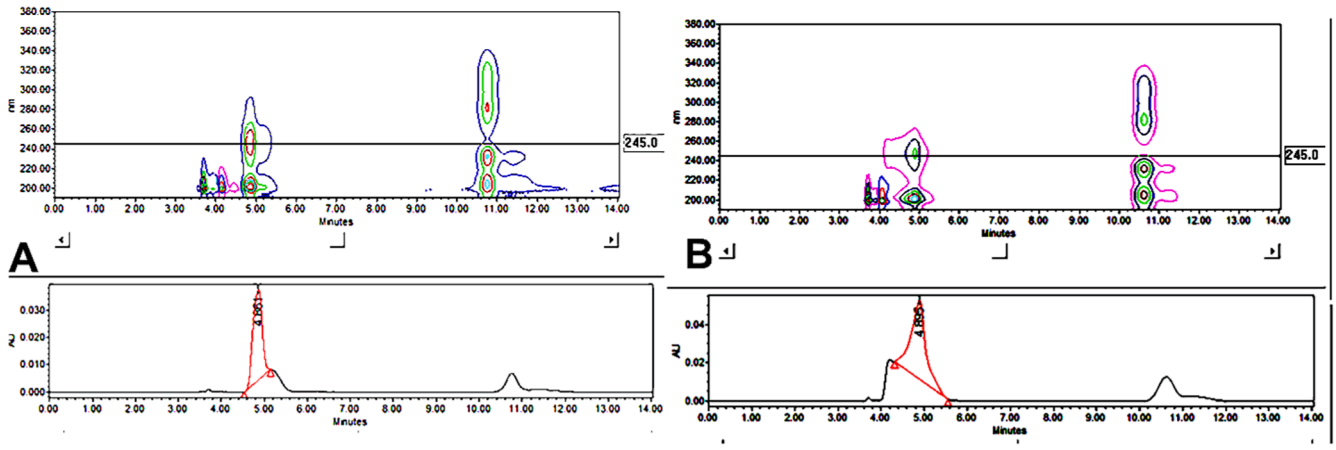


Fig. 12. HPLC chromatogram: Fresh Mixture (A) and One month Old Mixture (B).

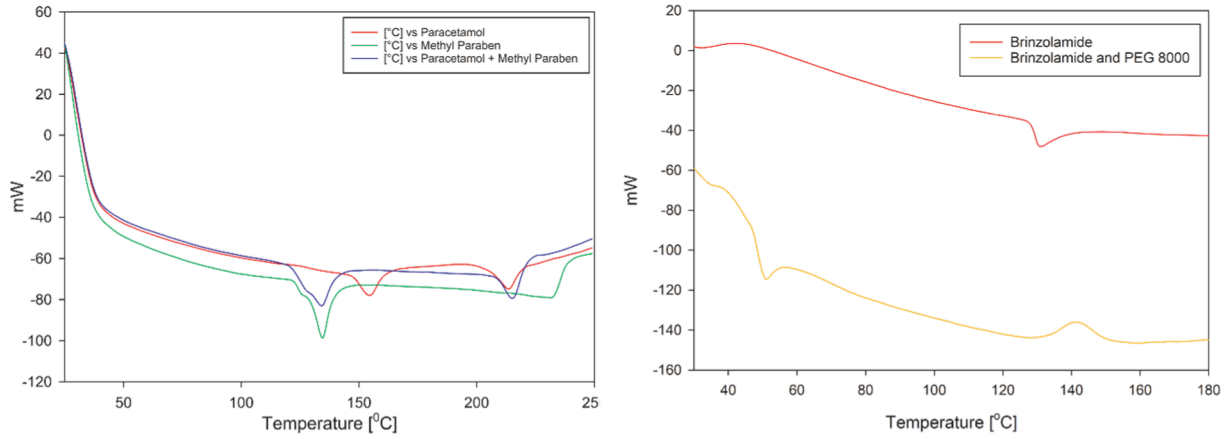


Fig. 13. DSC study of paracetamol, methylparaben, and a physical mixture thereof (Left), Brinzolamide and physical mixture of brinzolamide with PEG 8000 (Right).

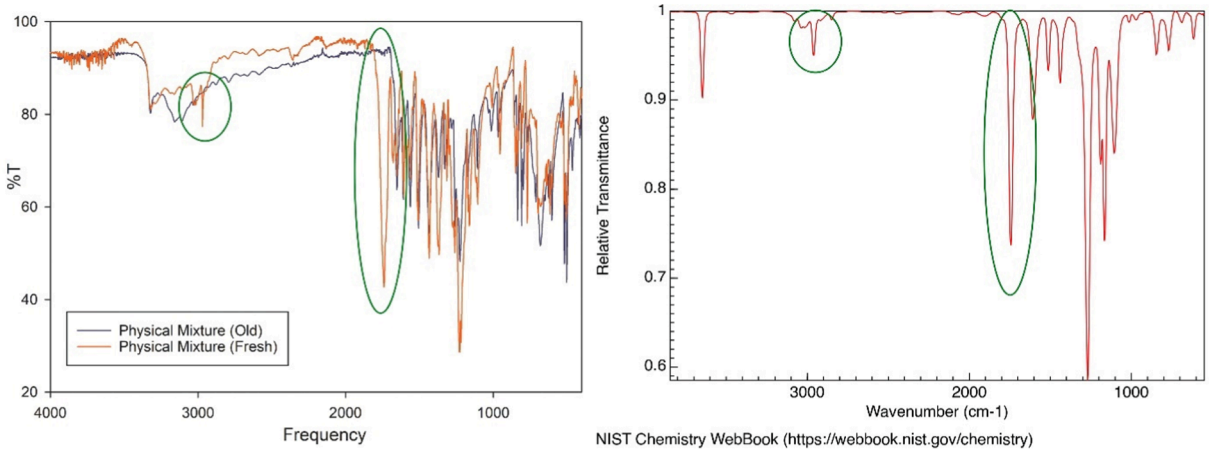


Fig. 14. FTIR study of PCM-Methyl Paraben physical mixture and Standard FTIR spectra of Methyl Paraben.

elsewhere, are listed in Table 7. Validation of these incompatibilities is out of the present article's scope and must be presented as separate work.

The PubChem fingerprint-based input for classifying the incompatible and compatible combinations of drug and excipient is well conceptualized in the present work. Though data are unbalanced, the model (DE-Interact) accurately predicts incompatible instances. Further eventual updates in the data can even reduce the chances of error. The machine learning models for getting the best accuracy, precision and loss are tried before finalizing the final version of the model. The model with maximum validation precision is finalized as a DE-Interact model for further predictions and actual validation through sophisticated analytical tools. The neural network developed with three hidden layers of 1024 nodes in each can understand the input of 1765 binary values to classify incompatible combinations precisely. Precision, Recall and F1 scores ratify utilizing the chemical fingerprint for incompatibility

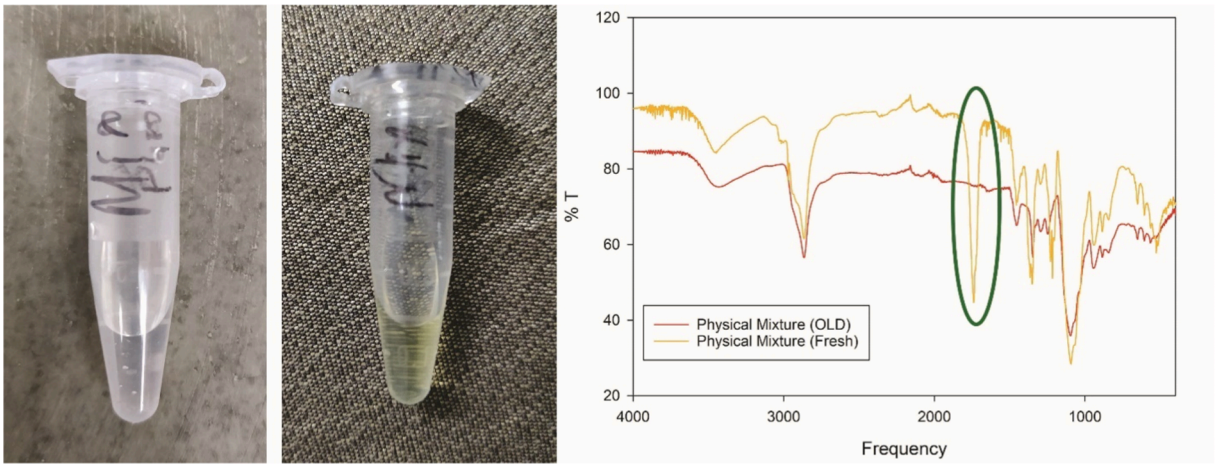


Fig. 15. Physical appearance of fresh (left) and old (right) mixture of brinzolamide and PEG 400 and FTIR study of both samples.

Table 7
Incompatible combinations of drug and excipients predicted by DE-Interact.

Drug	Excipient	Probability of Incompatibility by DE-Interact
Aceclofenac	Magnesium Stearate	0.6116
Promethazine	Lactose	0.9686
Temazepam	Dicalcium Phosphate Dihydrate	0.9999
Ketoprofen	Magnesium Stearate	0.9999
Famotidine	Dicalcium Phosphate Dihydrate	0.9999
Glipizide	Meglumine	0.9921
Enalapril	Microcrystalline Cellulose	0.9784

prediction.

The dataset compiled for the present work does not include just a chemical incompatibility. Both physical and chemical incompatibilities are collected from peer-reviewed research work and patented documents. Few incompatibilities, for instance, can reduce drug release eventually, change tablet hardness and disintegration time, induce phase inversion, etcetera. The DE-Interact model can suggest a possible incompatibility between two materials. Identifying physical incompatibility and validating the DE-Interact model's prediction is challenging. However, chemical incompatibilities can be easily observed through various analytical tools. The paracetamol–vanillin, paracetamol – methylparaben and brinzolamide–polyethyleneglycol is incompatible predictions by DE-Interact. DSC, FTIR, HPTLC, HPLC, and change in physical appearance are employed to validate the prediction of DE-Interact. Distortion in DSC and FTIR characteristic peaks are clear indications of chemical incompatibilities. Results can further be supported with HPTLC and HPLC analysis. In the case of brinzolamide and polyethyleneglycol, a literal colour change can be observed due to high reaction kinetics.

4. Conclusion

The outcome of the present work corroborates the notion of considering PubChem fingerprints as a descriptor in drug-excipient interaction. PubChem carries the entire chemistry of the compound in the dataset. The ANN model can identify the critical inputs from the fingerprint and correlate them with the binary outputs – compatible and incompatible. The DE-Interact model trained perfectly to predict the possibility of drug-excipient interactions. The model's accuracy may improve even more with the extension of the dataset. DE-Interact can undoubtedly be helpful to pharmaceutical formulators while choosing the excipients to avoid unnecessary stability-related issues. The physical validation of DE-Interact with three instances of the drug-excipient combinations in the current work is undoubtedly insufficient; nevertheless, the same can be carried out as a separate project with a specific focus on the performance validation of the model.

5. Declarations

Funding: Not Applicable.
Ethics approval: Not applicable.
Consent to participate: Not Applicable.
Consent for publication: Not Applicable.
Availability of data and material: Available from GitHub® and the link is mentioned in the introduction section.

CRediT authorship contribution statement

Swayamprakash Patel: Conceptualization, Methodology. **Mehul Patel:** . **Mangesh Kulkarni:** Data curation. **Mruduka S. Patel:** Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors are sincerely grateful to students – Ms. Mahima Wadje (M.Pharm), Ms. Bela Mehta, Ms. Priyanshi Marfatia, Ms. Prachi Patel and Ms. Janvi Patel (B.Pharm) for their assistance in collecting the dataset. These are the students of Ramanbhai Patel College of Pharmacy, Charotar University of Science and Technology.

References

Agatonovic-Kustrin, S., Beresford, R., 2000. Basic concepts of artificial neural network (ANN) modeling and its application in pharmaceutical research. J. Pharm. Biomed. Anal. 22, 717–727.
Akers, M.J., 2002. Excipient-drug interactions in parenteral formulations. J. Pharm. Sci. 91, 2283–2300.
Aysha, S., Suriya, S., Satyajit Roy, R., Fatema, M., Purabi, R.S., Hossain, M.d., S.,, 2019. Formulation and pharmacopoeial quality evaluation of ketorolac tromethamine IR tablet and comparison with marketed product. J. Appl. Pharm. Sci. 9, 82–87.
Bach, M., Werner, A., Żywiec, J., Pluskiewicz, W., 2017. The study of under-and over-sampling methods' utility in analysis of highly imbalanced data on osteoporosis. Inf. Sci. 384, 174–190.
Bajaj, S., Singla, D., Sakhuja, N., 2012. Stability testing of pharmaceutical products. J. Appl. Pharm. Sci. 129–138.

Balestrieri, F., Magrì, A.D., Magrì, A.L., Marini, D., Sacchini, A., 1996. Application of differential scanning calorimetry to the study of drug-excipient compatibility. Thermochim Acta 285, 337–345.
Burgina, E.B., Baltakhinov, V.P., Boldyreva, E.V., Shakhtschneider, T.P., 2004. IR Spectra of Paracetamol and Phenacetin. 1. Theoretical and Experimental Studies. J. Struct. Chem. 45, 64–73.
Chadha, R., Bhandari, S., 2014a. Drug-excipient compatibility screening–role of thermoanalytical and spectroscopic techniques. J. Pharm. Biomed. Anal. 87, 82–97.
Chadha, R., Bhandari, S., 2014b. Drug–excipient compatibility screening—role of thermoanalytical and spectroscopic techniques. J. Pharm. Biomed. Anal. 87, 82–97.
Chandrakala, V., Srinath, M.S., Saral, A.M., Kumar, S.U., 2011. Formulation and Evaluation of Bioadhesive Cyproheptadine Tablets. Trop. J. Pharm. Res. 10.
Chawla, N.V., 2010. Data Mining for Imbalanced Datasets: An Overview. In: Maimon, O., Rokach, L. (Eds.), Data Mining and Knowledge Discovery Handbook. Springer, US, Boston, MA, pp. 875–886.
Ertekin, Ş., 2013. Adaptive oversampling for imbalanced data classification, Information Sciences and Systems 2013. Springer 261–269.
Flach, P., Kull, M., 2015. Precision-recall-gain curves: PR analysis done right. Advances in neural information processing systems 28.
Gaikwad, S.S., Chafle, S.A., Morris, P.S., Avari, J.G., 2016. Development and evaluation of bilayer tablets of combination of antibiotics for the treatment of sexually transmitted disease. Braz. J. Pharm. Sci. 52, 555–566.
García-Arieta, A., 2014. Interactions between active pharmaceutical ingredients and excipients affecting bioavailability: impact on bioequivalence. Eur. J. Pharm. Sci. 65, 89–97.
Garreta, R., Moncecchi, G., 2013. Learning scikit-learn: machine learning in python. Packt Publishing Ltd.
Ghaferi, M., Amari, S., Mohrir, B.V., Raza, A., Shahmabadi, H.E., Alavi, S.E., 2020. Preparation, Characterization, and Evaluation of Cisplatin-Loaded Polybutylcyanoacrylate Nanoparticles with Improved In Vitro and In Vivo Anticancer Activities. Pharmaceuticals (Basel) 13.
González-González, O., Ramirez, I.O., Ramirez, B.I., O'Connell, P., Ballesteros, M.P., Torrado, J.J., Serrano, D.R., 2022. Drug Stability: ICH versus Accelerated Predictive Stability Studies. Pharmaceutics 14, 2324.
Gorain, B., Choudhury, H., Pandey, M., Madheswaran, T., Kesharwani, P., Tekade, R.K., 2018. Drug-Excipient Interaction and Incompatibilities. Dosage Form Design Parameters 363–402.
Grisoni, F., Consonni, V., Todeschini, R., 2018. Impact of Molecular Descriptors on Computational Models. Methods Mol. Biol. 1825, 171–209.
Gupta, N., 2013. Artificial neural network. Network Complex Syst. 3, 24–28.
Han, S.-H., Kim, K.W., Kim, S., Youn, Y.C., 2018. Artificial neural network: understanding the basic concepts without mathematics. Dementia Neurocogn. Disorders 17, 83.
Hassoun, M.H., 1995. Fundamentals of artificial neural networks. MIT press.
Israr, F., Mahmood, Z.A., Hassan, F., Hasan, S.M.F., Jabeen, S., Naz, S., Bashir, L., 2014. Formulation design and evaluation of Cefuroxime axetil 125 mg immediate release tablets using different concentration of sodium lauryl sulphate as solubility enhancer. Braz. J. Pharm. Sci. 50, 943–953.
Kumar, A., Sarkar, S., Pradhan, C., 2020. Malaria Disease Detection Using CNN Technique with SGD, RMSprop and ADAM Optimizers, Deep Learning Techniques for Biomedical and Health Informatics. Springer 211–230.
Labhsetwar, S.R., Haridas, S., Panmand, R., Deshpande, R., Kolte, P.A., Pati, S., 2020. Performance Analysis of Optimizers for Plant Disease Classification with Convolutional Neural Networks. arXiv preprint arXiv:2011.04056.
Lee, N., Yang, H., Yoo, H., 2021. A surrogate loss function for optimization of $ F_\beta $ score in binary classification with imbalanced data. arXiv preprint arXiv: 2104.01459.
Liang, D., 2014. Oral Liquid Formulation of Etravirine for Enhanced Bioavailability. J. Bioequivalence Bioavailability 06.
Maind, S.B., Wankar, P., 2014. Research paper on basic of artificial neural network. Int. J. Recent Innov. Trends Comput. Commun. 2, 96–100.
Mathew, J., Pang, C.K., Luo, M., Leong, W.H., 2017. Classification of imbalanced data by oversampling in kernel space of support vector machines. IEEE Trans. Neural Networks Learn. Syst. 29, 4065–4076.
Narang, A.S., Desai, D., Badawy, S., 2012. Impact of excipient interactions on solid dosage form stability. Pharm. Res. 29, 2660–2683.
Nivedithaa, V.R., Saba, M., 2018. Formulation and evaluation of immediate release combination tablet for cardiovascular diseases. Res. J. Life Sci., Bioinformatics, Pharm. Chem. Sci. 4.
Panakanti, R., Narang, A.S., 2012. Impact of excipient interactions on drug bioavailability from solid dosage forms. Pharm. Res. 29, 2639–2659.
Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., 2011. Scikit-learn: Machine learning in Python. J. Mach. Learn. Res. 12, 2825–2830.
Pokharana, M., Vaishnav, R., Goyal, A., Shrivastava, A., 2018. Stability testing guidelines of pharmaceutical products. J. Drug Deliv. Therapeutics 8, 169–175.
Remya, K., Beena, P., Bijesh, P., Sheeba, A., 2010. Formulation development, evaluation and comparative study of effects of super disintegrants in cefixime oral disintegrating tablets. J. Young Pharm. 2, 234–239.
Ruby, A.U., Prasannavenkatesan Theerthagiri, D.I., Vamsidhar, Y., 2020. Binary cross entropy with deep learning technique for Image classification. Int. J. 9.
Rumelhart, D.E., Widrow, B., Lehr, M.A., 1994. The basic ideas in neural networks. Commun. ACM 37, 87–93.
Shendryk, I., Rist, Y., Lucas, R., Thorburn, P., Ticehurst, C., 2018. Deep learning-a new approach for multi-label scene classification in planetscope and sentinel-2 imagery. In: IGARSS 2018–2018 IEEE International Geoscience and Remote Sensing Symposium. IEEE, pp. 1116–1119.
Swain, M., 2017. PubChemPy.
Thomas, V.H., Naath, M., 2008. Design and utilization of the drug-excipient chemical compatibility automated system. Int. J. Pharm. 359, 150–157.
Wakasawa, T., Sano, K., Hirakura, Y., Toyo'oka, T., Kitamura, S., 2008. Solid-state compatibility studies using a high-throughput and automated forced degradation system. Int. J. Pharm. 355, 164–173.
Xue, L., Bajorath, J., 2000. Molecular descriptors in chemoinformatics, computational combinatorial chemistry, and virtual screening. Comb. Chem. High Throughput Screen. 3, 363–372.
Yap, C.W., 2011. PaDEL-descriptor: an open source software to calculate molecular descriptors and fingerprints. J. Comput. Chem. 32, 1466–1474.
Yaqub, M., Jinchao, F., Zia, M.S., Arshid, K., Jia, K., Rehman, Z.U., Mehmood, A., 2020. State-of-the-Art CNN Optimizer for Brain Tumor Segmentation in Magnetic Resonance Images. Brain Sci. 10, 427.
Zarmpi, P., Flanagan, T., Meehan, E., Mann, J., Fotaki, N., 2020. Impact of Magnesium Stearate Presence and Variability on Drug Apparent Solubility Based on Drug Physicochemical Properties. Aaps J. 22, 75.
Zhou, L., Zhang, P., Chen, Z., Cai, S., Jing, T., Fan, H., Mo, F., Zhang, J., Lin, R., 2017. Preparation, characterization, and evaluation of amphotericin B-loaded MPEG-PCL-g-PEI micelles for local treatment of oral Candida albicans. Int. J. Nanomed. 12, 4269–4283.